ISB 140/DS 110−INTRODUCTION TO STATISTICS

LECTURE NOTES

Unit II: DESCRIPTIVE MEASURES

- ✓ One of the most important objectives of statistical analysis is to get one single value that describes the characteristic of the entire mass of data, that is by looking at descriptive measures.
- ✓ Ther are two types of descriptive measures and these include: **Measure of central tendency** and **Measure of dispersion**.

1. MEASURE OF CENTRAL TENDENCY (MEASURE OF LOCATION).

- ✓ Measures of central tendency are sometimes called averages, and they measure the location where most of the data set is clustered.
- ✓ There are three main statistical measures of central tendency or measure of location (or averages), these are arithmetic mean, median and mode.
- ✓ These averages can be computed for both ungrouped data as well as grouped data.
- ✓ Each of these measures has unique **properties**, along with advantages (**merits**) and disadvantages (**demerits**).
- ✓ **NOTE**
  1. A property is refers to a Fundamental Characteristic of a measure which describes how a measure behaves under certain conditions.
  2. A Merit (Advantage or Strength) is a benefit or useful feature of a measure that makes it preferable in certain situations. It answers the question: "Why is this measure useful?"
  3. A Demerit (Disadvantage or Weakness) is a limitation or drawback of a measure that affects its usefulness in some cases. It answers the question: "When does this measure fail?"

1.1. THE ARITHMETIC MEAN (OR AVERAGE)

- ✓ The **mean**, also called the **Arithmetic mean** is the most commonly used measure of centre.
- ✓ A **measure of centre** gives the centre of a histogram or a frequency distribution curve.
- ✓ The mean of a data set is its arithmetic average or arithmetic mean.
- ✓ It is the sum of the observations divided by the number of observations.

$$i.e., \text{Mean} = \frac{\text{Sum of observation}}{\text{Total number of observations}}$$

1.1.1. PROPERTIES

1. It is the sum of all observations divided by the total number of observations.

2. It is affected by extreme values (outliers).

   NOTE: An **outlier** is an extremely high or low value compared to the rest of the dataset. Since the mean considers all data points in its calculation, an outlier **pulls the mean toward itself**, making it unrepresentative of the central tendency of the dataset. For example,

   Consider the dataset: **5, 6, 7, 8, 9**

   $$\text{Mean} = \frac{5+6+7+8+9}{5} = \frac{35}{5} = 7$$

   Now, if we add an outlier (e.g., 50): **5, 6, 7, 8, 9, 50**

   Consider the dataset: **5, 6, 7, 8, 9**

   $$\text{Mean} = \frac{5+6+7+8+9+50}{5} = \frac{85}{6} = 14.17$$

   So, the mean jumps from 7 to 14.17, which is not representative of most values. Thus, To reduce the impact of outliers, alternatives like the **median** or **trimmed mean** (which removes extreme values before calculation) can be used.

3. It is unique for a given dataset.
4. It can be used in algebraic operations such as Sum of Deviations from the Mean which gives zero: $\sum(\mathrm{X}_i - \overline{\mathrm{X}}) = 0$, Mean of a Linear Transformation: $\overline{\mathrm{Y}} = \mathrm{a}\overline{\mathrm{X}} + \mathrm{b}$ etc.
5. $\bar{X}_i$ $(i = 1, 2, \ldots, k)$ are the means of k −component series of sizes n_i, $(i = 1, 2, \ldots, k)$ respectively, then the mean $\bar{X}$ of the composite series obtained on combining the component series is given by the formula:

$$\bar{X} = \frac{n_1\bar{X}_1 + n_2\bar{X}_2 + \cdots + n_k\bar{X}_k}{n_1 + n_2 + \cdots + n_k} = \frac{\sum_i n_i\bar{X}_i}{\sum_i n_i}$$

EXAMPLE

Tile average salary of male employees in a firm was k520 and those of males was k420. The mean salary of all the employees was k500. Find the percentage of male, and female employees.

Solution. Letn_1 and n_2 denote respectively the number of male and female employees in the concern and $\bar{X}_1$ and $\bar{X}_2$ denote respectively their average salary (in kwacha). Let $\bar{X}$ denote the average salary of all the workers in the firm.

We are given that:

$$\bar{X}_1 = 520, \qquad \bar{X}_2 = 420 \text{ and } \bar{X} = 500$$

Also we know

$$\bar{X} = \frac{n_1\bar{X}_1 + n_2\bar{X}_2}{n_1 + n_2}$$

$$\Longrightarrow 500 = \frac{520\, n_1 + 420\, n_2}{n_1 + n_2}$$

$$\Longrightarrow 500\,(n_1 + n_2) = 520\, n_1 + 420\, n_2$$

$$\Longrightarrow (520 - 500)\, n_1 = (500 - 420)\, n_2$$

$$\Longrightarrow 20\, n_1 = 80\, n_2$$

$$\Longrightarrow \frac{n_1}{n_2} = \frac{80}{20} \;\therefore\; \frac{n_1}{n_2} = \frac{4}{1}$$

Hence the percentage of male employees in the firm is:

$$\frac{4}{4+1} \times 100 = 80$$

and percentage of female employees in the firm is:

$$\frac{1}{4+1} \times 100 = 20$$

1.1.2. ADVANTAGES

1. It is the simplest to understand and the easiest to compute or calculate.
2. It Uses all data points, making it reliable for large datasets.
3. It is useful for further statistical analysis and inferences such as standard deviation estimation, variance and hypothesis testing.
1. It is Ideal for normally distributed data.
2. It is rigidly (or firmly) defined.

1.1.3. DISADVANTAGES

1. It is not always a good measure of central tendency, because it is highly sensitive to outliers, which can distort the true central value or it is affected by outlying observations (i.e., extreme values).
2. It cannot be calculated when the last class is open−ended (i.e., last class where upper boudary is not defined e.g 41 and above).
3. The Mean calculated may not be a data value in a data set.
4. It cannot be used for qualitative (or categorical) data.
5. The mean can be Misleading if the dataset is skewed.

✓ The following means are related to the Arithmetic mean:

A. Sample Mean

✓ The values of a variable in a sample from a population are called **sample data.**

✓ The mean of sample data is called a **sample mean**.

- ✓ It is the arithmetic average (mean) of sample data.
- ✓ In other words, the sample mean $(\bar{X})$ is the point estimator of the population mean (μ).
- ✓ The symbol used for a sample mean is a bar over the letter representing the variable.
- ✓ So, for a variable X, we denote a sample mean as $\bar{X}$ read as " X bar."
- ✓ If we also use the letter *n* to denote the sample size or, equivalently, the number of observations, then the definition of a sample mean can be expressed concisely.
- ✓ Symbolically, a sample mean for **un grouped data** is given by:

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} x_i\,, \qquad \text{where } n \text{ is a sample size.}$$

- ✓ Similarly, for **grouped data** the sample mean is given by:

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} fx_i\,, \text{where } n = \sum f \;\; and \;\; \sum(sigma) = summation$$

- ✓ In other words, Σx is the sum of all values and $\bar{X}$ is the sample mean.
- ✓ Thus, for **grouped data** the sample mean is also given as:

$$\bar{X} = \frac{\sum_{i=1}^{n} fx_i}{\sum f}$$

- ✓ However, the **population mean** is given by:

$$\mu = \frac{1}{N}\sum_{i=1}^{n} x_i,$$

where N is the number of observation in the population (or population size) and μ is the population.

EXAMPLES

1. Find the mean of the set of numbers given as follows:

63, 65, 67, 68, 69, 70, 71, 72, 74, 75.

SOLUTION

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} x_i = \frac{63 + 65 + 67 + \cdots + 75}{10} = \frac{694}{10} = 69.4$$

2. Use the frequency table below to compute the sample mean

No. of flowers (*X*)	**1**	**2**	**3**	**4**	**5**
No. of plants (*f*)	11	10	5	3	1

SOLUTION

X	f	fx
1	11	11
2	10	20
3	5	15
4	3	12
5	1	5
Total	$\sum f = 30$	$\sum fx = 63$

$$\bar{X} = \frac{\sum fx}{\sum f} = \frac{63}{30} = 2.1$$

3. The lengths of 32 leaves were measured correct to the nearest mm. Find the mean length.

Length	$20-22$	$23-25$	$26-28$	$29-31$	$32-34$
Frequency	3	6	12	9	2

SOLUTION

Length (mm)	Midpoint (x)	f	fx
$20-22$	21	3	63
$23-25$	34	6	144
$26-28$	27	12	324
$29-31$	30	9	270
$32-34$	33	2	66
Total		$\sum f = 32$	$\sum fx = 867$

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$$\therefore \bar{X} = \frac{\sum fx}{\sum f} = \frac{867}{32} = 27.1\ mm$$

B. TRIMMED MEAN

- ✓ we have earlier learned that the mean as a measure of centre is impacted by outliers.
- ✓ Thus, when a data set contains outliers, we can use either median or trimmed mean as a measure of the centre of a data set.
- ✓ Defn: Trimmed Mean is the mean obtained from the values remained after dropping $k\%$ of the values from each end of a ranked data set, where where k is any positive number,

such as 5%, 10%, and so on.

- ✓ Thus, to calculate the trimmed mean for a data set, we consider the following steps:

(1) Rank the given data in increasing order.

(2) Drop $k\%$ of the values from each end of the ranked data

(3) Then obtain the mean of the remaining values which is called the $k\%$ trimmed mean.

EXAMPLE

The following data give the money spent (in dollars) on books during 2015 by 10 students selected from a small college.

890, 1354, 1861, 1644, 87, 5403, 1429, 1993, 938, 2176

Calculate the 10% trimmed mean.

SOLUTION

To calculate the trimmed mean, first we rank the given data as shown below.

87, 890, 938, 1354, 1429, 1644, 1861, 1993, 2176, 5403

To calculate the 10% trimmed mean, we drop 10% of the data values from each end of the ranked data.

$10\%\ of\ 10\ \text{values}\ =\ 10\ (.10)\ =\ 1$

Hence, we drop one value from each end of the ranked data. After we drop the two values, one

from each end, we are left with the following eight values:

890, 938, 1354, 1429, 1644, 1861, 1993, 2176

The mean of these eight values will be called the 10% trimmed mean. Since there are 8 values,

$n\ =\ 8$. Adding these eight values, we obtain Σx as follows:

$$\Sigma x\ =\ 890\ +\ 938\ +\ 1354\ +\ 1429\ +\ 1644\ +\ 1861\ +\ 1993\ +\ 2176$$
$$=\ 12{,}285$$

The 10% trimmed mean will be obtained by dividing 12,285 by 8 as follows:

$$10\%\ \text{Trimmed Mean} = \frac{12{,}285}{8} = 1535.625\ =\ \$\mathbf{1535.63}$$

Thus, by dropping 10% of the values from each end of the ranked data for this example, we can state that students spent an average of $1535.63 on books in 2015.

Since in this data set $87 and $5403 can be considered outliers, it makes sense to drop these two values and calculate the trimmed mean for the remaining values rather than calculating the mean of all 10 values.

C. WEIGHTED MEANS

- ✓ In order to find the centre of the given data set especially where different values of a data set occur with different frequencies, or where each value of a data set is assigned different weight, we calculate the weighted mean.
- ✓ To calculate the weighted mean for a data set,

  (1) we denote the variable by *x* and the weights by *w*.

  (2) We add all the weights and denote this sum by Σw.

  (3) Then we multiply each value of *x* by the corresponding value of *w*.

  (4) The sum of the resulting products gives Σxw.

  (5) Thus, dividing Σxw by Σw gives the weighted mean.
- ✓ Symbolically, the weighted mean is calculated as:

$$\text{Weighted Mean} = \frac{\sum xw}{\sum w}$$

where *x* and *w* denote the variable and the weights, respectively.

EXAMPLE

Maura bought gas for her car four times during June 2024. She bought 10 gallons at a price of \$2.60 a gallon, 13 gallons at a price of \$2.80 a gallon, 8 gallons at a price of \$2.70 a gallon, and 15 gallons at a price of \$2.75 a gallon. What is the average price that Maura paid for gas during June 2024?

SOLUTION

Here the variable is the price of gas per gallon, and we will denote it by *x*. The weights are the number of gallons bought each time, and we will denote these weights by *w*. We list the values of *x* and *w* in the Table as below and find Σw. Then we multiply each value of *x* by the corresponding value of *w* and obtain Σxw by adding the resulting values. Finally, we divide Σxw by Σw to find the weighted mean.

Price in dollars (x)	Gallons of Gas (w)	xw
2.60	10	26.00
2.80	13	36.40
2.70	8	21.60
2.75	15	41.25
Total	$\sum w = 46$	$\sum xw = 125.25$

$$\text{Weighted Mean} = \frac{\sum xw}{\sum w} = \frac{125.25}{46} = \$2.72$$

Thus, Maura paid an average of \$2.72 a gallon for the gas she bought in June 2024.

1.1.4. RELATIONSHIPS BETWEEN ARITHMETIC MEAN, GEOMETRIC MEAN, AND HARMONIC MEAN.

- ✓ The **Arithmetic Mean (AM)**, **Geometric Mean (GM)**, and **Harmonic Mean (HM)** are three types of averages used in mathematics and statistics.
- ✓ Each has its own application, and they are related to each other in interesting ways, especially when dealing with **positive real numbers**.
- ✓ Here's a detailed breakdown of their definitions, relationships, and when to use them.

1. **Arithmetic Mean** (average) is given by

$$AM = \frac{x_1 + x_2 + \cdots + x_n}{n}$$

2. **Geometric Mean** (GM)

- ❖ Geometric Mean (GM) of the set of n observations is the nth root of their product.
- ❖ Thus, geometric Mean of the set of n observations $x_i, i = 1, 2, \ldots, n$ is given by

$$GM = (x_1.x_2.\ldots.x_n)^{1/n}$$

- ❖ The computation is facilitated by the use of logarithms. Thus, taking logarithm of both sides, we get

$$log\ (GM) = log\ (x_1.x_2.\ldots.x_n)^{1/n}$$
$$= \frac{1}{n} log\ (x_1.x_2.\ldots.x_n)$$

$$log(GM) = \frac{1}{n}(\log x_1 + logx_2 + \cdots + logx_n) = \frac{1}{n}\sum_{i=1}^{n} logx_i$$

$$\therefore \quad GM = Antilog\left[\frac{1}{n}(\log x_1 + logx_2 + \cdots + logx_n)\right]$$

$$= Antilog\left[\frac{1}{n}\sum_{i=1}^{n} logx_i\right]$$

- ❖ In case of frequency distribution $f_i, (\ i = 1, 2, \ldots, n)$ geometric mean, GM is given by

$$GM = \left(x_1^{f_1}.\ x_2^{f_2}.\ldots.x_n^{f_n}\right)^{1/N}, \quad where\ N = \sum_{i=1}^{n} f_i$$

- ❖ Taking logarithms of both sides, we get

$$log(GM) = \frac{1}{N}(f_1 \log x_1 + f_2 logx_2 + \cdots + f_n logx_n) = \frac{1}{N}\sum_{i=1}^{n} f_i logx_i$$

- ❖ Thus, we see that logarithm of GM is the arithmetic mean of the logarithms of the given values (**relationship**).

❖ Now taking the antilogarithm (e) both sides, we get

$$GM = Antilog\left[\frac{1}{N}\sum_{i=1}^{n} f_i log x_i\right]$$

❖ In the case of grouped or continuous frequency distribution, x is taken to be the value corresponding to the mid-point of the class-intervals.

RESULT

Suppose n_1 and n_2 are the sizes, GM_1 and GM_2 are the geometric means of two series respectively, then the geometric mean GM, of the combined series is given by

$$log(GM) = \frac{n_1 log(GM_1) + n_2 log(GM_2)}{n_1 + n_2}$$

ADVANTANGES

(i) It is rigidly defined.

(ii) It is based upon all the-observations

(iii) It is not affected much by fluctuations of sampling.

(vi) It gives comparatively more weight to small items.

DISADVANTAGES

(i) Because of its abstract mathematical character, geometric mean is not easy to understand and to calculate for a non-mathematics such a person.

(ii) If anyone of the observations is zero, geometric mean becomes zero and if anyone of the observations is negative geometric mean becomes imaginary regardless of the magnitude of the other items.

USES

(i) Geometric mean is used to find the rate of population growth and the rate of interest.

(ii) it is also used in the construction of index numbers.

3. **Harmonic Mean** (HM)

❖ Harmonic mean of a number of observations is the reciprocal of the arithmetic mean of the reciprocals of the given values.

❖ Thus, Harmonic mean (HM), of n observations x_i $(i = 1, 2, \dots, n)$ is given by:

$$HM = \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \cdots + \frac{1}{x_n}}$$

$$= \frac{n}{\frac{1}{n}\sum_{i=1}^{n} 1/x_i}$$

❖ In case of frequency distribution $f_i, (\ i = 1, 2, \ldots, n)$, harmonic mean, HM is given by

$$HM = \frac{N}{\frac{1}{f_1} + \frac{1}{f_2} + \cdots + \frac{1}{f_n}}$$

$$= \frac{N}{\frac{1}{N}\sum_{i=1}^{n} f_i/x_i}, \qquad where\ N = \sum_{i=1}^{n} f_i$$

ADVANTAGES

(i) Harmonic mean is rigidly defined,

(ii) It is based upon all the observations

(iii) It is suitable for further mathematical treatment.

DISADVANTAGES

(i) Like geometric mean, it is not affected much by fluctuations of sampling.

(ii) It gives greater importance to small items and

(iii) It is useful only when small items have to be given a greater weightage.

USES

It is used when averaging rates such as speed and efficiency etc.

EXAMPLE

A cyclist pedals from his house to his college at a speed of 10km/h
and back from the college to his house at 15km/h. Find the average speed?

SOLUTION

Solution. Let the distance from the house to the college be *x miles.* In going from house to college, the distance ($x\ miles$) is covered in $\frac{x}{10}$ hours, while in coming from college to house, the distance is covered $\frac{x}{15}$ hours. Thus, a total distance of $2x$ miles is covered in $\left(\frac{x}{10} + \frac{x}{15}\right)$ hours. Hence,

$$Average\ speed = \frac{total\ distance\ travelled}{Total\ time\ taken}$$

$$= \frac{2x}{\left(\frac{x}{10} + \frac{x}{15}\right)}$$

$$= \frac{2}{\left(\frac{1}{10} + \frac{1}{15}\right)}$$

$$= \frac{2}{\left(\frac{3+2}{30}\right)} = \frac{2}{\frac{5}{30}} = 12km/h$$

Thus, the average speed is given by the harmonic mean of 10 and 15 and not by the arithmetic mean.

NOTE

If we consider two positive numbers a and b, then

1. Arithmetic Mean is

$$AM = \frac{a+b}{2}$$

2. Geometric Mean is

$$GM = (a.b)^{1/2} = \sqrt{ab}$$

3. Harmonic Mean is

$$HM = \frac{2}{\frac{1}{a}+\frac{1}{b}} = \frac{2}{\frac{a+b}{ab}} = \frac{2ab}{a+b}$$

- ✓ In general, for any set of positive numbers, we have the relationship between Arithmetic mean, Geometric mean and Harmonic mean that:

$$AM \geq GM \geq HM$$

- ✓ Equality holds **if and only if** all the numbers are **equal**, i.e.,

$$x_1 = x_2 = \cdots = x_n$$

SUMMARY OF THE MEAN TYPE

Mean Type	Best Used For	Examples
Arithmetic Mean	General average when all values have equal weight	Average temperature, average score.
Geometric Mean	When dealing with **ratios**, **growth rates**, or **percentages**.	Average interest rate over time, population growth.
Harmonic Mean	When averaging rates (e.g., speed, efficiency).	Average speed over same distance, average cost per unit.

1.2. THE MEDIAN

- ✓ The median isone of the frequently used measure of centre beside the Arithmetic mean.
- ✓ The median of a data set is defined differently as follows:

(1) It is the number that divides the bottom 50% of the data from the top 50% or it is a value such that 50% of the observations are smaller and 50% of the observations are larger.

(1) It is the middle value of a given data set in its ordered list or the middle value when the data are arranged in increasing sequence or

(2) It is a positional average, and it is the value such that half the observations in the dataset
are larger than it and half are smaller than it.

(4) It is the central value after the observations are ranked according to size.

(5) It is described as the numerical value separating the higher half of a sample, a population or a probability distribution.

1.2.1. PROPERTIES

1. It is the middle value when data is arranged in ascending order.
2. It is not affected by extreme values.
3. It represents the 50th percentile (half of the data lies below and half above).
4. It is more suitable for skewed distributions.
5. In general, the median of a data set is the value in the middle when the data items are arranged in ascending order.

1.2.2. ADVANTAGES

1. It is rigidly defined.
2. The median is easily understood and is easy to compute or calculate.
3. It is not at all affected by extreme values (outliers), making it ideal for skewed distributions.
4. It can be calculated for distributions with open-end classes and even when the data is qualitative rather than quantitative.
5. Suitable for ordinal data (e.g., ranks, satisfaction levels).
6. Always represents an actual data point (if the number of values is odd).

1.2.3. DISADVANTAGES

1. The median requires that observations be arranged in ascending order.
2. It does not use all the observations. Implying that the median does not consider all data points in calculations.
3. In case of even number of observations median cannot be determined exactly. We, merely estimate it by taking the mean of two middle terms.
4. As compared with mean, the median cannot be used for further statistical analysis like variance and standard deviation because it is affected much by fluctuations of sampling.
5. It is not useful for grouped data without interpolation.

1.2.4. USES

(i) Median is the only average to be used while dealing with qualitative data which cannot be measured quantitatively but still can be arranged in ascending or descending order of magnitude, e.g., to find the average intelligence or average honesty among a group of people.

(ii) It is to be used for determining the typical value in problems concerning wages, distribution of wealth, etc.

1.3. THE MODE

- ✓ First and fore most, we consider the following statements related to the mode:

  (i) The average height of a Zambian (male) is $5-8$.

  (ii) The average size of the shoes sold in a shop is 7.

  (iii) An average student in a hostel spends k500 per month.

- ✓ In all the above cases, the average referred to is mode.
- ✓ Thus, just like the median, the mode or modal value of a given data is also defined differently as follows:

  (1) It is the value which has maximum frequency or any value of the data which occurs repeatedly.

  (2) It is the observation with a maximum frequency.

  (3) It is the value that has the highest frequency in a data set.

  (4) It is the value that occurs with the greatest frequency.

- ✓ In general, Mode is the value which occurs most frequently in a set of observations and around which the other items of the set cluster densely.
- ✓ In other words, mode is the value of the variable which is predominant in the series.

NOTE

1. The greatest frequency can occur at two or more different value
2. If the data have exactly one mode, the data is unimodal.
3. If the data have exactly two modes, then the data is bimodal
4. If the data have more than two modes, then the data is multimodal.
5. In a grouped frequency distribution, the modal class is the interval with the greatest frequency density, that is, the interval represented by highest bar in the histogram**.**

- ✓ like the average Mean, the Median and the Mode can be computed for both ungrouped and grouped data.

1.3.1. PROPERTIES

1. Mode is the most frequently occurring value in a dataset.
2. A dataset can be unimodal (one mode), bimodal (two modes), or multimodal

(multiple modes).

3. It can be used for both numerical and categorical data.
4. It is not affected by extreme values.

1.3.2. ADVANTAGES

(1) Mode is readily comprehensible (clear) and easy to calculate (compute) and easy to Interpret.

(2) Like median, mode can be located in some cases merely by inspection

(3) It is not at all affected or influenced by the extreme values (outliers).

(4) It can be calculated even when the last class is open-ended or when the data are qualitative rather than quantitative.

(5) Mode can be conveniently located even if the frequency distribution has class-intervals of unequal magnitude provided the modal class and the classes preceding and succeeding it are of the same magnitude.

1.3.3. DISADVANTAGES

(1) Mode does not use all the information available or it is not based upon all the observations.

(2) Mode may not exist or there may be more than one mode.

(3) Mode is not always unique, implying that it is not always possible to find a clearly defined mode, because a dataset can have multimodal, bi−modal or no modes.

(4) It does not always provide a central value if the dataset has extreme variations.

(5) It is not useful for further statistical analysis.

1.3.4. USES

- ✓ Mode is the average to be used to find the ideal size, e.g., in business, forecasting, in the manufacture of ready-made garments, shoes. etc.

COMPARISON OF MEASURES OF CENTRAL TENDENCY

Measure	Best use when	Affected by outliers?	Usable for Categorical Data?	Used in Further Analysis
Mean	Symmetric data, no outliers	**Yes**	**No**	**yes**
Median	Skewed data, outliers present	**No**	**No**	**No**
Mode	Categorical or discrete data	**No**	**Yes**	**No**

1.4. MEDIAN FOR UNGROUPED (DISCRETE) AND GOUPED (CONTINUOUS) DATA

1.4.1. MEDIAN FOR UNGROUPED (DISCRETE) DATA

- ✓ If the number of observations is an odd number, then the median is the middle value of the observations.
- ✓ In other words, it is the observation that is exactly in the middle of the ordered list.
- ✓ If the number of observations is an even number, then the median is the average of the two middle or central values or it is the mean of the two middle observations in the ordered list.
- ✓ To determine the median of ungrouped data, arrange the data in increasing or ascending order, then Cancel the data values uniformly from the left and the right of the given data set.
- ✓ The midle value or values remaining is the median, if two or more values remain at the middle, we find the average of the values.
- ✓ Thus, if we let the ordered values of a data set be $X_1, X_2, \dots, X_n$, where *n* is the number of observations, then the median of the ungrouped data denoted by M_d is given by:

$$M_d = \begin{cases} X_{\frac{n+1}{2}}, & \text{if } n \text{ is odd} \\ \frac{1}{2}\left(X_{\frac{n}{2}} + X_{\frac{n+2}{2}}\right), & \text{if } n \text{ is even} \end{cases}$$

EXAMPLE

Find the Median of the following data set:

(a) 2, 1, 0, 3, 5, 2, 3, 1, 0, 2, 2,2, 4 (b) 1, 1, 2, 3, 0, 2, 4.

SOLUTION

(a) 0, 0, 1, 1, 2, 2, ②, 2, 2, 3, 3, 4, 5 $[n = 13 \text{ (odd number)}]$ or

$$Median\ (M_d) = X_{\frac{n+1}{2}} = X_{\frac{13+1}{2}} = X_{\frac{14}{2}} = X_7 = 2$$

Interpretation: The 7th data value is the median.

(b) 0,1, 1, 2, 2, 3, $[n = 6 \text{ (even number)}]$, so the Median is given by

$$\text{Median } (M_d) = \frac{1}{2}\left(X_{\frac{n}{2}} + X_{\frac{n+2}{2}}\right) = \frac{1}{2}\left(X_{\frac{6}{2}} + X_{\frac{6+2}{2}}\right) = \frac{1}{2}(X_3 + X_4) = \frac{1+2}{2}$$

$$= 1.5$$

REMARK

In case of discrete frequency distribution, median is obtained by considering the cumulative frequencies.

The steps for calculating median for discrete frequency distribution are given below:

(i) Find $\frac{N}{2}$, where $N = \sum_i f_i$

(ii) See the (less than) cumulative frequency $(cf.)$ just greater than $\frac{N}{2}$

(iii) The corresponding value of *x* is median.

EXAMPLE

Obtain the median for the following frequency distribution:

x	1	2	3	4	5	6	7	8	9
f	8	10	11	16	20	25	15	9	6

SOLUTION

x	f	$c.f$
1	8	8
2	10	18
3	11	29
4	16	45
5	20	65
6	25	90
7	15	105
8	9	114
9	6	**120**
	$N = 120$	

Hence $N = 120 => N/2 = 120/2 = 60$. Cumulative frequency $(cf.)$ just greater than $N/2 = 60$, is 65 and the value of *x* corresponding to 65 is 5. Therefore, median is 5.

1.4.2. MEDIAN FOR GROUPED (CONTINUOUS) DATA

✓ For any grouped data or any continuous frequency distribution, the class corresponding to the cf just greater than $N/2$ is called the **median class** and the value of median is obtained by the following formula:

$$\text{Median}(M_d) = L_m + \left(\frac{\frac{N}{2} - CF}{f_m}\right) i$$

where, $L_m =$ lower class boudary of the median class

$CF =$ cumulative frequency of a class immediately before median class

$f_m =$ frequency of the median class

$i =$ median class width

✓ The median for grouped data is determined by considering the following steps:

Step 1: Construct the cumulative frequency distribution.

Step 2: Decide or locate the class median also known as the median class.

The Median Class is the first class that contains $\frac{N^{th}}{2}$ observation, where N is the total frequency.

Step 3: Find the median by using the above following formula:

EXAMPLES

1. Based on the grouped data below, find the median

Time to travel to work	**Frequency**
$\mathbf{1-10}$	8
$\mathbf{11-20}$	14
$\mathbf{21-30}$	12
$\mathbf{31-40}$	9
$\mathbf{41-50}$	7

SOLUTION

Time to travel to work	Class boundary	Frequency (f)	Cumulative frequency (CF)
$1-10$	$0.5-10.5$	8	8
$11-20$	$10.5-20.5$	14	22
$21-30$	$20.5-30.5$	12	34
$31-40$	$30.5-40.5$	9	43
$41-50$	$40.5-50.5$	7	**50**
Total		$N=\mathbf{50}$	

The median class contains $(N/2)^{th} = (50/2)^{th}$ observation, that is, the 25th observation and it is contained in the class $21-30$, so the median class is the $3rd$ class, $CF = 22$, $f_m = 12$, $L_m = 20.5$ and $i = 10$.

Alternatively: Here $N/2 = 50/2 = 25$. Cumulative frequency just greater than 25 is 34 and the corresponding class is $21-30$. Thus median class is $21-30$.

$$\therefore \text{Median}(M_d) = L_m + \left(\frac{\frac{N}{2} - CF}{f_m}\right) i$$

$$= 20.5 + \left(\frac{25-22}{12}\right)10 = 23.$$

2. The frequency table below shows the salaries that 30 employees in a certain firm received in a month.

Amount	N0. of employees
$1-10$	3
$11-20$	7
$21-30$	11
$31-40$	5
$41-50$	4
Total	30

Calculate the median.

SOLUTION

Amount	Class boundary	Frequency (f)	Cumulative frequency (CF)
$1-10$	$0.5-10.5$	3	3
$11-20$	$10.5-20.5$	7	10
$21-30$	$20.5-30.5$	11	21
$31-40$	$30.5-40.5$	5	26
$41-50$	$40.5-50.5$	4	**30**
Total		$N = \mathbf{30}$	

The median class contains $(N/2)^{th} = (30/2)^{th}$ observation, that is, the 15th observation and it is contained in the class $21-30$, so the median class is the $3rd$ class, $CF = 10$, $f_m = 11$, $L_m = 20.5$ and $i = 10$.

Alternatively: Here $N/2 = 30/2 = 15$. Cumulative frequency just greater than 15 is 21 and the corresponding class is $21-30$. Thus median class is $21-30$.

$$\therefore \text{Median}(M_d) = L_m + \left(\frac{\frac{N}{2} - CF}{f_m}\right)i$$

$$= 20.5 + \left(\frac{15-10}{11}\right)10$$

$$= 25.05$$

3. The grouped data in the Table below represent the number of children from birth through the end of the teenage years in a large apartment complex. Find the median.

Class	$0-3$	$4-7$	$8-11$	$12-15$	$16-19$
Frequency	7	4	19	12	8

SOLUTION

Class	Class boundary	Frequency (f)	Cumulative frequency (CF)
$0-3$	$-0.5-3.5$	7	7
$4-7$	$3.5-7.5$	4	11
$8-11$	$7.5-11.5$	19	30
$12-15$	$11.5-15.5$	12	42
$16-19$	$15.5-19.5$	8	**50**
Total		$N=\mathbf{50}$	

The median class contains $(N/2)^{th}=(50/2)^{th}$ observation, that is, the 25th observation and it is contained in the class $8-11$, so the median class is the $3rd$ class, $CF=11$, $f_m=19$, $L_m=7.5$ and $i=4$.

Alternatively: Here $N/2=\ 50/2=\ 25$. Cumulative frequency just greater than 25 is 30 and the corresponding class is $8-11$. Thus median class is $8-11$.

$$\therefore \text{Median}(\ M_d\ )=L_m+\left(\frac{\frac{N}{2}-CF}{f_m}\right)i$$

$$=7.5+\left(\frac{25-11}{19}\right)4$$

$$=10.45$$

4. Find the median wage of the following distribution:

Wages (k)	$20-30$	$30-40$	$40-50$	$50-60$	$60-70$
No. of Labourers	3	5	20	10	5

SOLUTION

Wages (k)	No. of Labourers (f)	Cumulative frequency (CF)
$20-30$	3	3
$30-40$	5	8
$40-50$	20	28
$50-60$	10	38
$60-70$	5	**43**
Total	$N = \mathbf{43}$	

The median class contains $(N/2)^{th} = (43/2)^{th}$ observation, that is, the 21.5^{th} observation and it is contained in the class $40-50$, so the median class is the $3rd$ class, $CF = 8$, $f_m = 20$, $L_m = 40$ and $i = 10$.

Alternatively: Here $N/2 = 43/2 = 21.5$. Cumulative frequency just greater than 21.5 is 28 and the corresponding class is $40-50$. Thus median class is $40-50$.

$$\therefore \text{Median}(M_d) = L_m + \left(\frac{\frac{N}{2} - CF}{f_m}\right) i$$

$$= 40 + \left(\frac{21.5 - 8}{20}\right) 10$$

$$= 46.75$$

Thus, median wage is k46.75

1.5. THE GRAPHICAL LOCATION OF THE MEDIAN

✓ The **graphical location of the median** refers to how the median of a dataset is visually represented on a graph, such as a box plot, histogram, or cumulative frequency graph. Let's break it down:

1.5.1. BOX PLOT (BOX-AND-WHISKER PLOT)

✓ The median is represented by the **central line inside the box**.

✓ The box plot divides the dataset into **quartiles**, where:

(i) The lower quartile also known as the first quartile (Q_1) marks the 25th percentile,

(ii) The median also known as the second quartile (Q_2) marks the 50th percentile,

(iii)The upper quartile also known as the third quartile (Q_3) marks the 75th percentile.

NOTE

Further discusion on the boxplot and quartile shall be done in the next session.

1.5.2. HISTOGRAM

- ✓ The median is the point on the $x - axis$ where half the data is to the left and half to the right**.**
- ✓ It may or may not align with the peak (mode) of the histogram.
- ✓ If the data is symmetrical, the median is at the center.
- ✓ If the data is skewed, the median shifts toward the longer tail.

1.5.3. CUMULATIVE FREQUENCY GRAPH

- ✓ The median is located at the 50th percentile (cumulative frequency = 50%)**.**
- ✓ To find it graphically, a horizontal line is drawn from the 50% mark on the $y - axis$ to the cumulative frequency curve**,** and then a vertical line is dropped to the $x - axis$.

1.5.4. IMPORTANCE OF THE GRAPHICAL LOCATION OF THE MEDIAN

- ✓ It helps in identifying **data symmetry or skewness**.
- ✓ It shows the **central tendency** of the dataset.
- ✓ It is useful in **comparing multiple datasets** visually.

1.6. MODE FOR UNGROUPED (DISCTETE) AND GROUPED (CONTINUOUS) DATA

1.6.1. MODE FOR UNGROUPED DATA

- ✓ For ungrouped or discrete frequency distribution, mode is the value x corresponding to maximum frequency.
- ✓ In other words, mode for discrete data can be found easily by inspection.

EXAMPLE

Find the Mode of the following data set:

(a) 2, 1, 0, 3, 5, 2, 3, 1, 0, 2, 2,2, 4 (b) 1, 1, 2, 3, 0, 2, 4.

SOLUTION

(a) 0, 0, 1, 1, 2, 2, 2,②, 2, 3, 3, 4, 5. The mode is 2 (the most appearing number).

(b) 0,1, 1, 2, 2, 3, 4

Value	0	1	2	3	4
Frequency	1	2	2	1	1

There are two mode values in the data set, that is, 1 and 2 (bimodal) having high frequency.

NOTE

For anyone (or more) of the following cases:

(i) If the maximum frequency is repeated,

(ii) If the maximum frequency occurs in the very beginning or at the end of the distribution, and

(iii) If there are irregularities in the distribution, the value of mode is determined by the method of grouping.

EXAMPLE

Find the mode of the following frequency distribution

Size (x)	1	2	3	4	5	6	7	8	9	10	11	12
Frequency (f)	3	8	15	23	35	40	32	28	20	45	14	6

SOLUTION

Here we see that the distribution is not regular since the frequencies are increasing steadily up to 40 and then decrease but the frequency 45 after 20 does not seem to be consistent with the distribution. So, we cannot say that since maximum frequency is 45, mode is 10. We shall locate mode by the method of grouping as explained below:

4. The frequencies in column (i) are the original frequencies.
5. Column (ii) is obtained by combining the frequencies two by two.
6. If we leave the first frequency and combine the remaining frequencies two by two, we get column (iii).
7. Combining the frequencies two by two after leaving the first two frequencies results in a repetition of column (ii).
8. Hence, we proceed to combine the frequencies three by three, thus getting column (iv).
9. The combination of frequencies three by three after leaving the first frequency results in column (v) and after leaving the first two frequencies results in column (vi).

Size	*Frequency*					
(x)	*(i)*	*(ii)*	*(iii)*	*(iv)*	*(v)*	*(vi)*
1	3					
		11				
2	8		**23**			
				26		
3	15				**46**	
		38				**73**
4	23		**58**			
				98		
5	35					
		75			**107**	
6	40		**72**			**100**
7	32					
		60		**80**		
8	28		**48**			
					93	
9	20					**79**
		65	**59**			
10	**45**			**65**		
11	14					
		20				
12	6					

The maximum frequency in each column is given in bold b1ack type. To find mode

we form the following table:

Column Number (1)	*Maximum Frequency (2)*	*Value or combination of values of x giving max. frequency in (2) (3)*
(i)	45	10
(ii)	75	5, 6
(iii)	72	6, 7
(iv)	98	4, 5, 6,
(v)	107	5, 6, 7
(vi)	100	6, 7, 8

On examining the values in column (3) above. we find that the value 6 is repeated the maximum number of times and hence the value of mode is 6 and not 10 which is an irregular item.

1.6.2. MODE FOR GROUPED DATA (CONTINOUS DATA)

- ✓ When the data are grouped, the modal class is the class with the highest frequency.
- ✓ To find the mode for grouped data, we use the following formula:

$$Mode(\ M_o\ ) = L_{m_o} + \left(\frac{\Delta_1}{\Delta_1 + \Delta_2}\right) i$$

where, L_{m_o} = lower class boudary of the modal class

Δ_1= Difference between the frequency of the modal class and the frequency of the class before the class mode.

Δ_2= Difference between the frequency of the modal class and the frequency of the class after the class mode.

i = class width

EXAMPLES

1. Based on the grouped data below, find the mode.

Time to travel to work	Frequency
$1 - 10$	8
$11 - 20$	14
$21 - 30$	12
$31 - 40$	9
$41 - 50$	7

SOLUTION

Based on the table, by inspection the highest frequency is 14, and so the modal class is $11 - 20$ and L_{mo} is the lower limit of the modal class.

$L_{mo} = 10.5, \Delta_1 = 14 - 8 = 6, \Delta_2 = 14 - 12 = 2$ and $i = 10$

$$Mode = L_{mo} + \left(\frac{\Delta_1}{\Delta_1 + \Delta_2}\right) i$$

$$= 10.5 + \left(\frac{6}{6+2}\right)10 = 18$$

2. The frequency table below shows the salaries that 30 employees in a certain firm received in a month.

Amount	N0. of employees
$1 - 10$	3
$11 - 20$	7
$21 - 30$	11
$31 - 40$	5
$41 - 50$	4
Total	30

Calculate the mode.

SOLUTION

Based on the table, by inspection the highest frequency is 11, and so the modal class is $21 - 30$ and L_{mo} is the lower limit of the modal class.

$L_{mo} = 20.5, \Delta_1 = 11 - 7 = 4, \Delta_2 = 11 - 5 = 6$ and $i = 10$

$$Mode = L_{mo} + \left(\frac{\Delta_1}{\Delta_1 + \Delta_2}\right) i$$

$$= 20.5 + \left(\frac{4}{4+6}\right)10 = 24.5$$

3. The grouped data in the Table below represent the number of children from birth through the end of the teenage years in a large apartment complex. Find the mode.

Class	$0 - 3$	$4 - 7$	$8 - 11$	$12 - 15$	$16 - 19$
Frequency	7	4	19	12	8

SOLUTION

Based on the table, by inspection the highest frequency is 19, and so the modal class is $8 - 11$ and L_{mo} is the lower limit of the modal class.

$L_{mo} = 7.5, \Delta_1 = 19 - 4 = 15, \Delta_2 = 19 - 12 = 7$ and $i = 4$

$$Mode = L_{mo} + \left(\frac{\Delta_1}{\Delta_1 + \Delta_2}\right) i$$

$$= 7.5 + \left(\frac{15}{15+7}\right)4 = 10.23$$

REMARK

- ✓ When the modal class (the class with the highest frequency) is at one end of the distribution, in this case, the last class or the first class, you still use the mode-formula for grouped data but must handle the fact that there is no following class.
- ✓ One common approach is to assume that the frequency in the next (non-existent) class is zero.
- ✓ Then use the standard mode formula
- ✓ We get the width from the two proceeding classes.

EXAMPLES

1. The grouped data in the Table below represent the number of children from birth through the end of the teenage years in a large apartment complex. Find the mode.

Class	$0-3$	$4-7$	$8-11$	$12-15$	$16-19$
Frequency	7	4	8	12	19

SOLUTION

Based on the table, by inspection the highest frequency is 19, and so the modal class is $8-11$ and L_{mo} is the lower limit of the modal class.

$L_{mo} = 15.5, \Delta_1 = 19-12 = 7, \Delta_2 = 19-0 = 19$ and $i = 4$

$$Mode = L_{mo} + \left(\frac{\Delta_1}{\Delta_1 + \Delta_2}\right) i$$

$$= 15.5 + \left(\frac{7}{19+7}\right) 4 = 16.58$$

2. Consider the following grouped data below, find the mode.

(a)

Class interval	**Frequency**
$\mathbf{0-10}$	30 ← modal class
$\mathbf{10-20}$	18
$\mathbf{20-30}$	10
$\mathbf{30-40}$	5

$L_{mo} = 0, \Delta_1 = 30-0 = 30, \Delta_2 = 30-18 = 12$ and $i = 10$

$$Mode = L_{mo} + \left(\frac{\Delta_1}{\Delta_1 + \Delta_2}\right) i$$

$$= 0 + \left(\frac{30}{30+12}\right) 10 = 7.14$$

(b)

Class Interval	Frequency
$\mathbf{0-10}$	6
$\mathbf{10-20}$	12
$\mathbf{20-30}$	18
$\mathbf{30-40}$	25← modal class

$L_{mo} = 30, \Delta_1 = 25 - 18 = 7, \Delta_2 = 25 - 0 = 25$ and $i = 10$

$$Mode = L_{mo} + \left(\frac{\Delta_1}{\Delta_1 + \Delta_2}\right) i$$

$$= 30 + \left(\frac{7}{7 + 25}\right) 10 = 32.19$$

REMARK

✓ When you have two or more modal classes, we estimate the mode by considering the modal class as the one with the steeper increase and decrease in frequencies around it.

EXAMPLE

Class Interval	Frequency
$\mathbf{10-20}$	8
$\mathbf{20-30}$	12← Modal class 1
$\mathbf{30-40}$	12← Modal class 2
$\mathbf{40-50}$	7

SOLUTION

Here: The Highest $frequency = 12$ and two adjacent classes have the same frequency

Now, to compute the mode, **we choose the class that has a steeper increase from the previous class** or a steeper fall into the next class.

Let's analyse:

- Increase from $10-20$ to $20-30$: $12 - 8 = +4$
- Increase from $20-30$ to $30-40$: $12 - 12 = 0$
- Decrease from $30-40$ to $40-50$: $12 - 7 = +5$

So, $\mathbf{30-40}$ has a steeper drop after it, hence, we will treat **30–40** as the **modal class**.

$L_{mo} = 30, \Delta_1 = 12 - 12 = 0, \Delta_2 = 12 - 7 = 5$ and $i = 10$

$$Mode = L_{mo} + \left(\frac{\Delta_1}{\Delta_1 + \Delta_2}\right) i$$

$$= 30 + \left(\frac{0}{0 + 5}\right) 10 = 30$$

2. MEASURE OF DISPERSION OR MEASURE OF VARIATION

- ✓ It is often desirable to consider measures of variability (dispersion), as well as measure of location.
- ✓ Measures of variability or variation include the following: Range, Variance, Standard Deviation etc.

2.1. RANGE

- ✓ The range of a data set is the difference between the largest and smallest data values or it is the difference between highest and lowest values in the data set or distribution.
- ✓ The most common application of the range is in statistical quality control.
- ✓ Thus, range, is given by:

$$Range = Largest\ value - Smallest\ value$$

- ✓ Fore example, the Range of the sample:

53, 55, 70, 58, 64, 57, 53, 69, 57, 68, 53 is

$$Range = Largest\ value - Smallest\ value = 70 - 53 = 17$$

ADVANTAGE

(i) It is the simplest measure of variability

(ii) It is very sensitive to the smallest and largest data values.

DISADVANTAGE

(i) Range is a very poor measure of the variability since it involves only two numbers.

(ii) It provides no indication about the spread of the other values.

2.2. VARIANCE AND STANDARD DEVIATION

- ✓ The **variance** and **standard deviation** are the most used measures of variability among the observations.
- ✓ The variance is a measure of variability that utilizes all the data.
- ✓ It is the average of the squared differences between each data value and the mean.
- ✓ In other words, variance is the mean squared deviation of the values from the mean.
- ✓ The standard deviation of a data set is the positive square root of the variance.
- ✓ It is measured in the same units as the data, making it more easily interpreted than the variance.
- ✓ The variance and standard deviation can be computed for both ungrouped and grouped data as follows:

2.2.1. FOR UNGROUPED DATA

✓ If $X_1, X_2, \dots, X_N$ are the *N* observations in the population, then

1. the population variance, denoted by σ^2 is given by:

$$\sigma^2 = \frac{(X_1-\mu)^2 + (X_2-\mu)^2 + \cdots + (X_N-\mu)^2}{N} = \frac{1}{N}\sum_{i=1}^{N}(X_i-\mu)^2$$

2. The population standard deviation, denoted by σ is given by:

$$\sigma = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(X_i-\mu)^2} = \sqrt{\sigma^2}$$

✓ Now, if we have *n* sample observations $x_1, x_2, \dots, x_n$, then

3. The sample variance, denoted by S^2 is given by:

$$S^2 = \frac{(x_1-\bar{x})^2 + (x_2-\bar{x})^2 + \cdots + (x_n-\bar{x})^2}{n-1} = \frac{1}{n-1}\sum_{i=1}^{n}(x_i-\bar{x})^2$$

$$\Leftrightarrow S^2 = \frac{1}{n-1}\left[\sum_{i=1}^{n}x_i^2 - \frac{(\sum x_i)^2}{n}\right]$$

4. As for the population standard deviation, the sample standard deviation denoted by *S* is given by:

$$S = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i-\bar{x})^2} = \sqrt{\frac{1}{n-1}\left[\sum_{i=1}^{n}x_i^2 - \frac{(\sum x_i)^2}{n}\right]} = \sqrt{S^2}$$

NOTE

S^2 has (*n* - 1) instead of *n* as a denominator, because by using *n* - 1, we get a better estimate for σ^2.

2.2.2. FOR GROUPED DATA

1. The population Variance is given by:

$$\sigma^2 = \frac{1}{N}\left[\sum_{i=1}^{N}fX_i^2 - \frac{(\sum fX_i)^2}{N}\right] = \frac{1}{\sum f}\left[\sum_{i=1}^{N}fX_i^2 - \frac{(\sum fX_i)^2}{\sum f}\right]$$

2. The population standard deviation is given by:

$$\sigma = \sqrt{\frac{1}{N}\left[\sum_{i=1}^{N}fX_i^2 - \frac{(\sum fX_i)^2}{N}\right]} = \sqrt{\frac{1}{\sum f}\left[\sum_{i=1}^{N}fX_i^2 - \frac{(\sum fX_i)^2}{\sum f}\right]} = \sqrt{\sigma^2}$$

3. The sample variance is given by

$$S^2 = \frac{1}{\sum f - 1}\sum_{i=1}^{n}f(x_i-\bar{x})^2 = \frac{1}{\sum f - 1}\left[\sum_{i=1}^{n}fx_i^2 - \frac{(\sum fx_i)^2}{\sum f}\right]$$

4. The sample standard deviation is given by

$$S = \sqrt{\frac{1}{\sum f - 1}\left[\sum_{i=1}^{n} f x_i^2 - \frac{\left(\sum f x_i\right)^2}{\sum f}\right]} = \sqrt{S^2}$$

EXAMPLES

1. Find the mean, variance and the standard deviation of the following scores 3, 5, 7, 9,11 and 13.

SOLUTION

$$Mean: \quad \bar{x} = \frac{1}{n}\sum_{i=1}^{n}(x_i)$$

$$= \frac{5 + 7 + 9 + 11 + 13}{6} = 8$$

$$Variance: \quad S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

$$= \frac{(3-8)^2+(5-8)^2+(7-8)^2+(9-8)^2+(11-8)^2+(13-8)^2}{5}$$

$$= \frac{69}{5}$$

$$= \mathbf{13.8}$$

$$Standard\ deviation: \quad SD = \sqrt{S^2} = \sqrt{13.8} = 3.71$$

2. Find the variance and standard deviation for the following data:

No.of Order	Frequency
$10 - 12$	4
$13 - 15$	12
$16 - 18$	20
$19 - 21$	14
Total	$\sum f = 50$

SOLUTION

1. $Variance: \quad S^2 = \frac{1}{\sum f - 1}\left[\sum_{i=1}^{n} f x_i^2 - \frac{\left(\sum f x_i\right)^2}{\sum f}\right]$

2. $Standard\ deviation\colon\ S = \sqrt{\frac{1}{\sum f - 1}\left[\sum_{i=1}^{n} f{x_i}^2 - \frac{\left(\sum f x_i\right)^2}{\sum f}\right]} = \sqrt{S^2}$

No.of Order	Frequency	x	fx	fx^2
$10-12$	4	11	44	484
$13-15$	12	14	168	2352
$16-18$	20	17	340	5780
$19-21$	14	20	280	5600
Total	$\sum f = 50$		$\sum fx = 832$	$\sum f\,x^2 = 14216$

$$Variance\colon\quad S^2 = \frac{1}{\sum f - 1}\left[\sum_{i=1}^{N} fx^2 - \frac{\left(\sum fx\right)^2}{\sum f}\right]$$

$$= \frac{1}{50-1}\left[14216 - \frac{832^2}{50}\right] = 7.5820$$

$$Standard\ deviation\colon S = \sqrt{S^2} = \sqrt{7.5820} = 2.75$$

3. The grouped data in the table below represent the number of children from birth through the end of the teenage years in a large apartment complex. Find the variance, and standard deviation for these data

Class	$0-3$	$4-7$	$8-11$	$12-15$	$16-19$
Frequency	7	4	19	12	8

SOLUTION

No.of Order	Frequency	x	fx	fx^2
$0-3$	7	1.5	10.5	15.75
$4-7$	4	5.5	22	121
$8-11$	19	9.5	180.5	1714.75
$12-15$	12	13.5	162	2187
$16-19$	8	17.5	140	2450
Total	$\sum f = 50$		$\sum fx = 515$	$\sum f\,x^2 = 6488.5$

$$Variance\colon\quad S^2 = \frac{1}{\sum f - 1}\left[\sum_{i=1}^{N} fx^2 - \frac{\left(\sum fx\right)^2}{\sum f}\right]$$

$$= \frac{1}{50-1}\left[16488.5 - \frac{515^2}{50}\right] = \frac{11184}{49} = 228.24$$

$$Standard\ deviation: S = \sqrt{S^2} = \sqrt{228.24} = 15.11$$

3. RELATIVE MEASURES OF DISPERSION (VARIABILITY)

- ✓ A relative measure of dispersion occurs when comparing the variability or dispersion of two or more data sets expressed in different units or in the same units, but different order of magnitude.
- ✓ The relatuve measures of variability include the coefficients of variations, quartiles and percentiles.
- ✓ However, the quartiles and percentiles are measures of relative position.

3.1. COEFFICIENTS OF VARIATION (CV)

- ✓ C.V is one of the relative measures of dispersion or variability.
- ✓ The coefficient of variation indicates how large the standard deviation is in relation to the mean.
- ✓ The coefficient of variation is computed as follows:
- ✓ For a sample data values it is given as :

$C.V = \left(\frac{S}{\bar{x}} \times 100\right)$, where S is a sample standard deviation and $\bar{x}$ *is* the sample mean.

- ✓ For a population data values it is given as:

$C.V = \frac{\sigma}{\mu} \times 100$, where σ is the population standard deviation and μ *is* the population mean.

NOTE

1. The series having greater $C.V$ is said to be more variable than the other and the series having lesser $C.V$ is said to be more consistent (or homogeneous) than the other.
2. A **measure of position** determines the position of a single value in relation to other values in a sample or a population data set.

EXAMPLES

1. A Lusaka based firm would like to purchase stock in one of two companies (A and B) listed on the Lusaka Stock exchange. The firm considered the monthly returns of the two companies over the last 10 months

A: 34 42 36 38 45 40 32 34 39 41

B: 21 24 32 64 50 35 28 30 42 55

Compare the variability in returns between the two companies. In which company should the firm invest?

SOLUTION

For company A

$$Mean\ (\bar{x}) = \frac{1}{n}\sum_{i=1}^{10}(x_i) = \frac{34+42+36+38+45+40+32+34+39+41}{10}$$

$$= \mathbf{38.1}$$

$$S = \sqrt{\frac{1}{n-1}\left[\sum_{i=1}^{10} x_i^2 - \frac{(\sum x_i)^2}{n}\right]}$$

$$\sum x_i = 34+42+36+38+45+40+32+34+39+41 = 381$$

$$\sum {x_i}^2 = 34^2+42^2+36^2+38^2+45^2+40^2+32^2+34^2+39^2+41^2$$

$$= 14667$$

$$\therefore\ S = \sqrt{\frac{1}{10-1}\left[14667 - \frac{381^2}{10}\right]} = 4.09$$

Thus,

$$C.V = \left(\frac{S}{\bar{x}} \times 100\right) = \frac{4.09}{38.1} \times 100 = 10.7\%$$

For company B

$$Mean\ (\bar{x}) = \frac{1}{n}\sum_{i=1}^{10}(x_i) = \frac{24+32+64+\cdots+42+55}{10}$$

$$= \mathbf{38.1}$$

$$\sum x_i = 21+24+\cdots+42+55 = 381$$

$$\sum {x_i}^2 = 21^2+24^2+\cdots+42^2+55^2 = 16335$$

$$\therefore\ S = \sqrt{\frac{1}{10-1}\left[16335 - \frac{381^2}{10}\right]} = 3.77$$

Thus,

$$C.V = \left(\frac{S}{\bar{x}} \times 100\right) = \frac{3.77}{38.1} \times 100 = 9.9\%$$

The returns of company A are more viable and therefore risky compared to company B. The Lusaka based firm should invest in company B.

2. From the given data in the table below compute the coefficients of variation for the twenty-five year old and for the eleven-year old students.

	Sample 1	Sample 2
Age	25 years	11 years
Mean weight	145 pounds	80 pounds
Standard deviation	10 pounds	10 pounds

SOLUTION

We have for twenty-five-year-old

$$C.V = \left(\frac{S}{\bar{x}} \times 100\right) = \frac{10}{145} \times 100 = 6.9\%$$

And for eleven-year-old

$$C.V = \left(\frac{S}{\bar{x}} \times 100\right) = \frac{10}{80} \times 100 = 12.5\%$$

TASK

John is deciding which insurance company he would like to buy his life assurance between company A and B. The following are the amounts in millions of Kwacha paid out in life assurance claims for both companies over 8 months in the 2024.

A:35 26 41 38 36 40 29 30

B:28 24 40 60 21 27 30 22

Calculate the coefficient of variation for each set of claims. Which company is more consistent in paying out claims? Justify your answer.

3.2. THE QUARTILES, PERCENTILES AND DECILES

- ✓ Qurtiles, percentiles and deciles can be calculated for both ungrouped and grouped data.
- ✓ How ever, the concentration in this course is to discuss quartiles and percentiles.
- ✓ Defn: Quartiles are summary measures that divide a ranked dataset into four equal parts (quarters).
- ✓ Defn: Percentiles indicate specific percent positions of data values within the dataset.
- ✓ In other words, percentiles divide the dataset into 100 equal parts.
- ✓ Defn: Deciles divide the data set into 10 equal parts.
- ✓ There are three types of quartile, these are:

  1.The **first quartile** also known as **Lower Quartile** denoted by (Q_1)

  2. The **second quartile** also called **Median** denoted by (Q_2)

3.The **third quartile** also called **Upper Quartile** denoted by (Q_3)

3.2.1. QUARTILES AND PERCENTILES FOR UNGROUPED DATA.

- ✓ For ungrouped data, quartiles and percentiles are found by calculating the position.
- ✓ This is similar to the calculations used for determining the position of the median for ungrouped data.
- ✓ In order to estimate the quartiles or percentiles, the data need to be ordered in sequence from lowest to highest rank.

Formula: Quartiles and percentiles for ungrouped data:

$$Quartile\ position = \frac{q(n+1)}{4}(round\ down)$$

$$Percentile\ position = \frac{p(n+1)}{100}(round\ down)$$

Where,

q is the required quartile, i.e., $1, 2$ or 3.

n is the sample size

p is the required percentile, i.e., $25th$ for 25% or $75th$ for 75%

NOTE

1. For a data set with n data values, we have the following positions of the Quartiles, i.e., Q_1, Q_2 and Q_3:

(a) The lower quartile (Q_1) position is given by

$$Q_1 = \frac{n+1}{4}\ if\ n\ is\ odd\ or\ Q_1 = \frac{n}{4}\ if\ \ n\ is\ even$$

(b)The median (Q_2) position is given by

$$Q_2 = \frac{n+1}{2}\ if\ n\ is\ odd\ or\ Q_2 = \frac{n}{2}\ if\ \ n\ is\ even$$

(c)The upper quartile (Q_3) position is given by

$$Q_3 = \frac{3(n+1)}{4}\ if\ n\ is\ odd\ or\ Q_1 = \frac{3n}{4}\ if\ \ n\ is\ even$$

2. Quartiles are specific percentiles. For example,

(a) $First\ Quartile\ (Q_1) = 25th\ Percentile\ (P_{25}) = \frac{25}{100} \times n = \frac{n}{4}$

(b) $Second\ Quartile\ (Q_2) = 50th\ Percentile (P_{50}) = \frac{50}{100} \times n = \frac{n}{2}$

(c) $Third\ quartile\ (Q_3) = 75th\ percentile (P_{75}) = \frac{75}{100} \times n = \frac{3n}{4}$

✓ Thus, the table below summarizes the rank or positions of quartiles for ungrouped (discrete) and grouped (continuous) data where n is the number of observation.

Quartiles	Rank when n is odd	Rank when n is even
Q_1	$(n+1)/4$	$n/4$
Q_2	$(n+1)/2$	$n/2$
Q_3	$3(n+1)/4$	$3n/4$

3.2.2. INTERQUARTILE RANGE (IQR)

✓ The Interquartile Range (IQR) of a data set is the difference between the third quartile (upper quartile) and the first quartile (lower quartile).

✓ It is the range for the middle 50% of the data, and it overcomes the sensitivity to extreme data values.

✓ Thus, the Interquartile Range (IQR) is given by

$$IQR = Q_3 - Q_1.$$

3.2.3 QUARTILE DEVIATION

✓ The quartile deviation is half the interquartile range of a dataset.

✓ It is a measure of the spread through the middle half of the dataset.

✓ It can be useful because it is not influenced by extremely high or extremely low values.

$$quartile\ deviation = \frac{IQR}{2} = \frac{Q_3 - Q_1}{2}$$

Where

$$IQR = Interquartile\ range$$

$$Q_3 = upper\ quartile$$

$$Q_1 = lower\ quartile$$

EXAMPLES

1. 40 pay TV subscribers have rated their experience (out of 100)

42	43	44	45	45	47	50	51
51	51	51	52	53	54	54	56
57	58	58	59	62	62	62	63
63	64	65	67	67	69	69	72
73	74	75	78	78	83	84	87

Calculate all quartiles, the 60th and 70th percentiles and the IQR of the dataset.

SOLUTION

The data need to be ranked in order to determine these values. This dataset has already been sequenced.

The positions for the required measures are then calculated, after which the position is pinpointed and the value determined.

$$first\ quartile\ position = \frac{1(n+1)}{4} = \frac{1(40+1)}{4} = 10.25 = 10\ (rounded)$$

$$The\ value\ of\ Q_3 = 51\ rating$$

$$second\ quartile\ position = \frac{2(n+1)}{4} = \frac{2(40+1)}{4} = 20.5 = 20\ (rounded)$$

$$The\ value\ of\ Q_2 = 59\ rating$$

$$third\ quartile\ position = \frac{3(n+1)}{4} = \frac{3(40+1)}{4} = 30.75 = 30(rounded)$$

$$The\ value\ of\ Q_3 = 69\ rating$$

Alternatively, since n is even, then

$$The\ position\ of\ Q_1 = \frac{n}{4} = \frac{40}{4} = 10$$

$$\therefore The\ value\ of\ Q_1 = 51\ rating$$

$$The\ position\ of\ Q_2 = \frac{n}{2} = \frac{40}{2} = 20$$

$$\therefore The\ value\ of\ Q_2 = 59\ rating$$

$$The\ position\ of\ Q_3 = \frac{3n}{4} = \frac{3 \times 40}{4} = 30$$

$$\therefore The\ value\ of\ Q_3 = 69\ rating$$

$$60th\ percentile\ position = \frac{60(n+1)}{100} = \frac{60(40+1)}{100} = 24.6 = 24\ (rounded)$$

$$\therefore The\ value\ of\ P_{60} = 63\ rating$$

$$70th\ percentile\ position = \frac{70(n+1)}{100} = \frac{70(40+1)}{100} = 28.7 = 28\ (rounded)$$

$$\therefore The\ value\ of\ P_{70} = 67\ rating$$

Alternatively

$$P_{60}\ position = \frac{60}{100} \times n = \frac{60}{100} \times 40 = 24$$

$$\therefore The\ value\ of\ P_{60} = 63\ rating$$

$$P_{70}\ position = \frac{70}{100} \times n = \frac{70}{100} \times 40 = 28$$

$$\therefore The\ value\ of\ P_{70} = 67\ rating$$

$$IQR = Q_3 - Q_1 = 69 - 51 = 18$$

2. Based on the sample data below, show that the 25th and 75th percentiles are equal to the first and third quartiles, respectively.

53, 55, 70, 58, 64, 57, 53, 69, 57, 68, 53

SOLUTION

53, 53, 53, 55, 57, 57, 58, 64, 69, 70 $\quad n = 11\ (odd)$

$$25th\ percentile\ position = \frac{25(n+1)}{100} = \frac{25(11+1)}{100} = 3$$

$$\therefore The\ value\ of\ P_{25} = 53$$

$$75th\ percentile\ position = \frac{75(n+1)}{100} = \frac{75(11+1)}{100} = 9$$

$$\therefore The\ value\ of\ P_{75} = 68$$

$$The\ position\ of\ Q_1 = \frac{n+1}{4} = \frac{11+1}{4} = 3$$

$$\therefore The\ value\ of\ Q_1 = 53$$

$$The\ position\ of\ Q_3 = \frac{3(n+1)}{4} = \frac{3 \times 12}{4} = 9$$

$$\therefore The\ value\ of\ Q_3 = 68$$

Thus, $P_{25} = Q_1 = 53\ and\ P_{75} = Q_3 = 68$, hence shown.

3.2.4. QUARTILES AND PERCENTILES FOR GROUPED DATA.

- ✓ Quartiles can be obtained from grouped data just as it is done in the median.
- ✓ For any grouped data or any continuous frequency distribution, the class corresponding to the cf just greater than $N/4$ or $3N/4$ is called the **lower quartile or upper quartile class**.
- ✓ Thus, the following computational formulae are then made use of to estimate the lower (Q_1) and upper (Q_3) quartiles:

$$Q_1 = L_{Q_1} + \left(\frac{\frac{N}{4} - CF}{f_{Q_1}}\right) i$$

where, L_{Q_1} = lower class boudary of the lower quartile class

CF = cumulative frequency of a class immediately before the $c.f$ of the lower quartile class

f_{Q_1} = frequency of the lower quartile class

i = lower quartile class width

$$Q_3 = L_{Q_3} + \left(\frac{\frac{3N}{4} - CF}{f_{Q_3}}\right) i$$

where, L_{Q_3} = upper class boudary of the upper quartile class

CF = cumulative frequency of a class immediately before the $c.f$ of the uppe quartile class

f_{Q_3} = frequency of the upper quartile class

$i = upper\ quartile$ class width

- ✓ In general, for grouped data the quartiles are given by:

$$Q_j = L_{Q_j} + \left(\frac{\frac{j \times N}{4} - CF}{f_{Q_j}}\right) i$$

- ✓ Where

j = is the required quartile $i.e., 1, 2\ or\ 3$

$\frac{j \times N}{4} = is\ the\ quartile\ position, \qquad N = \sum f$

- ✓ Similarly, the **percentiles** for any grouped data or any continuous frequency distribution can be determined using the formula below:

$$P_j = L_{P_j} + \left(\frac{\frac{j \times N}{100} - CF}{f_{P_{60}}}\right) i$$

✓ where,

$L_{P_j} = lower$ class boudary of the percentile class

$CF =$ cumulative frequency of a class immediately

before the $c.f$ of the percentile class

$f_{P_j} =$ frequency of the percentile class

$i = percentile$ class width

$j =$ is the required percentile $i.e., 25\%, 50\%$ or 75%

$\frac{j \times N}{100} =$ is the percentile position that determines the percentile class ,

$N = \sum f$

EXAMPLES

1. Based on the grouped data below, find the lower quartile, upper quartile and the quartile deviation.

Time to travel to work	**Frequency**
$\mathbf{1-10}$	8
$\mathbf{11-20}$	14
$\mathbf{21-30}$	12
$\mathbf{31-40}$	9
$\mathbf{41-50}$	7

SOLUTION

Time to travel to work	Class boundary	Frequency (f)	Cumulative frequency (CF)
$1-10$	$0.5-10.5$	8	8
$11-20$	$10.5-20.5$	14	22
$21-30$	$20.5-30.5$	12	34
$31-40$	$30.5-40.5$	9	43
$41-50$	$40.5-50.5$	7	**50**
Total		$N = \mathbf{50}$	

The lower quartile class contains $(N/4)^{th} = (50/4)^{th}$ observation, that is, the 12.5^{th} observation and it is contained in the class $11-20$, so the lower quartile class is the 2nd class, $CF = 8$, $f_{Q_1} = 14$, $L_m = 10.5$ and $i = 10$.

Alternatively: Here $N/4 = 50/4 = 12.5$. Cumulative frequency just greater than 12.5 is 22 and the corresponding class is $11 - 20$. Thus, lower quartile class is $11 - 20$.

$$\therefore\ Q_1 = L_{Q_1} + \left(\frac{\frac{N}{4} - CF}{f_{Q_1}}\right) i$$

$$= 10.5 + \left(\frac{12.5 - 8}{14}\right) 10 = 13.71.$$

Similarly, the upper quartile class contains $(3\,N/4)^{th} = (3 \times 50/4)^{th}$ observation, that is, the 37.5^{th} observation and it is contained in the class $31 - 40$, so the upper quartile class is the 4th class, $CF = 34$, $f_{Q_3} = 9$, $L_{Q_3} = 30.5$ and $i = 10$.

Alternatively: Here $3N/4 = 3 \times 50/4 = 37.5$. Cumulative frequency just greater than 37.5 is 43 and the corresponding class is $31 - 40$. Thus, upper quartile class is $31 - 40$.

$$\therefore\ Q_3 = L_{Q_3} + \left(\frac{\frac{3N}{4} - CF}{f_{Q_3}}\right) i$$

$$= 30.5 + \left(\frac{37.5 - 34}{9}\right) 10 = 34.39.$$

$$\text{Thus, } quartile\ deviation = \frac{Q_3 - Q_1}{2} = \frac{34.39 - 13.71}{2} = 10.34$$

2. Using the cumulative frequency distribution table below, calculate all quartiles and the 60th and 70th percentiles.

Interval (weight in kg)	Frequency (f)	Cumulative frequency (CF)
$140 - 150$	1	1
$150 - 160$	4	5
$160 - 170$	8	13
$170 - 180$	7	20
$180 - 190$	5	**25**
Total	$N = \mathbf{25}$	

SOLUTION

$Q_1\ position = (N/4)^{th} = (25/4)^{th} = 6.25\ th\ value$

$Q_1\ class = 160 - 170,$

$$\therefore\ Q_1 = L_{Q_1} + \left(\frac{\frac{N}{4} - CF}{f_{Q_1}}\right) i$$

$$= 160 + \left(\frac{6.25 - 5}{8}\right)10 = 161.56kg$$

$Q_2\ position = (N/2)^{th} = (25/2)^{th} = 12.5\ th\ value$

$Q_2\ class = 160 - 170,$

$$\therefore\ Q_2\ = L_{Q_2} + \left(\frac{\frac{N}{2} - CF}{f_{Q_2}}\right)i$$

$$= 160 + \left(\frac{12.5 - 5}{8}\right)10 = 169.38kg$$

$Q_3\ position = (3N/4)^{th} = (3 \times 25/4)^{th} = 18.75\ th\ value$

$Q_3\ class = 170 - 180,$

$$\therefore\ Q_3\ = L_{Q_3} + \left(\frac{\frac{3N}{4} - CF}{f_{Q_3}}\right)i$$

$$= 170 + \left(\frac{18.75 - 13}{7}\right)10 = 178.21kg$$

$60th\ percentile\ position = (j \times N/100)^{th} = (60 \times 25/100)^{th} = 15\ th\ value.$

$P_{60}\ class = 170 - 180,$

$$\therefore\ P_{60}\ = L_{P_{60}} + \left(\frac{\frac{j \times N}{100} - CF}{f_{P_{60}}}\right)i$$

$$= 170 + \left(\frac{15 - 13}{7}\right)10 = 172.86kg$$

$70th\ percentile\ position = (j \times N/100)^{th} = (70 \times 25/100)^{th} = 17.5th\ value.$

$P_{70}\ class = 170 - 180,$

$$\therefore\ P_{60}\ = L_{P_{60}} + \left(\frac{\frac{j \times N}{100} - CF}{f_{P_{60}}}\right)i$$

$$= 170 + \left(\frac{17.5 - 13}{7}\right)10 = 176.43kg$$

TASK

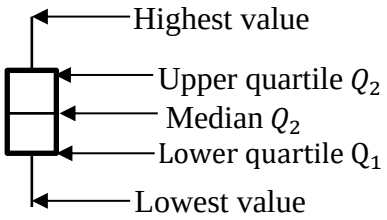
The time taken to complete an assembling task has been measured for 250 employees: Using the cumulative frequency table below, calculate all the quartiles and show that they are the specific percentiles.

Time taken (mins)	Number of people (f)	Cumulative frequency (CF)
$0-5$	2	2
$5-10$	2	4
$10-15$	3	7
$15-20$	5	12
$20-25$	5	17
$25-30$	18	35
$30-35$	85	120
$35-40$	92	212
$40-45$	37	249
$45-50$	1	**250**
Total	$N = \mathbf{250}$	

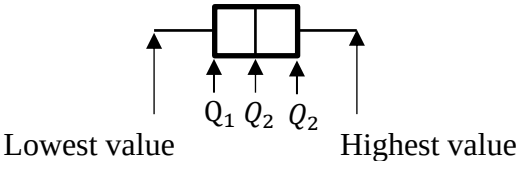
3.3. BOX AND WHISKER PLOTS OR BOX PLOT

- ✓ A boxplot (also called a box and whisker plot) can be used to display numerical features of data.
- ✓ In other words, a box-and-whisker plot is a graphical display in which a box extending from $Q_1$to Q_2 is constructed and contains the middle 50% of the data.
- ✓ The lines, called whiskers, are drawn from Q_1 to the smallest value and from Q_3 to the largest value.
- ✓ In addition, a vertical line constructed inside the box relates to the median.
- ✓ This plot can also be used to identify outliers (extreme or unusual observations).
- ✓ the box and whisker diagram or box plot illustrates the dispersion or spread of the distribution.
- ✓ The whiskers extend from the box to the highest and lowest values and illustrate the range of data.
- ✓ It uses the higest and the lowest values of the data, the quartiles (Q_2, Q_2 and Q_3) as shown below

Vertically



Horizontally



- ✓ Thus, to construct a box and whisker plot we follow the following steps:

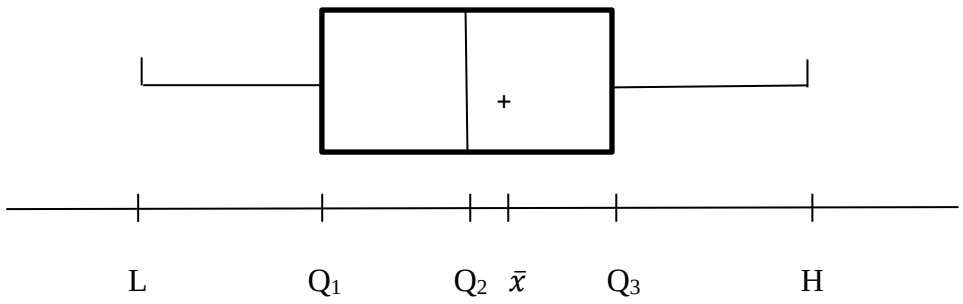
  Step1: Rank the observations from lowest (L) to highest (H).

  Step 2: Calculate Q_1(lower quartile), Q_2 (median), Q_3 (upper quartile) and the mean $(\bar{x})$.

  Step 3: Calculate $Q_1 - 1.5IQR$, if an observation falls below this value then it is an outlier.

  Step 4: Calculate $Q_3 + 1.5IQR$, if an observation lies above this value then it is an outlier. $(IQR = Interquartile\ range = Q_3 - Q_1)$.

- ✓ Thus, an outlier is defined as any observation less than $Q_1 - 1.5\,IQR$ or greater than $Q_3 + 1.5\,IQR$.
- ✓ If there are no outliers, then the boxplot takes the form:



- ✓ **NOTE**

  1. We start by drawing a box from the Q_1 to Q_3, dividing the box at Q_2 and then connecting the lowest and the highest observations to the box, provided the lowest and highest observations are not outliers.
  2. If there are outliers, the lowest and highest observations that are not outliers will be connected to the box and the outliers will be marked.
  3. Asterisks (stars) are used to denote outliers.
  4. Note also that the mean given by $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$ is denoted by + sign in the box plot to show its location.

- ✓ **EXAMPLES**

  1. Construct a boxplot for the following data

     27 32 47 54 36 44 56 49 53 51 35 29 62 19 65 37 48 33 41 45

  SOLUTION

  Ordering the observations (can be done using a stem and leaf plot. so we have:

  19 27 29 32 33 35 36 37 41 44 45 47 48 49 51 53 54 56 62 65

Method1

$$Q_1\ position = \frac{1(n+1)}{4} = \frac{1(20+1)}{4} = 5.25 = 5\ (round\ down)\ \text{or}$$

$$The\ position\ of\ Q_1 = \frac{n}{4} = \frac{20}{4} = 5, \qquad n = 20\ (even)$$

$$The\ value\ of\ Q_1 = 33$$

$$Q_2\ position = \frac{2(n+1)}{4} = \frac{(20+1)}{2} = 10.5 = 10\ (round\ down)\ \text{or}$$

$$The\ position\ of\ Q_2 = \frac{n}{4} = \frac{20}{4} = 10,\ n = 20\ (even)\ \text{or}$$

$$Q_2\ position = \frac{1}{2}\left(X_{\frac{n}{2}} + X_{\frac{n+2}{2}}\right) = \frac{1}{2}\left(X_{10} + X_{11}\right) = \frac{44+45}{2} = 44.5$$

$$= 44\ (rounded)$$

$$The\ value\ of\ Q_2 = 44$$

$$Q_3\ position = \frac{3(n+1)}{4} = \frac{3(20+1)}{4} = 15.75 = 15\ (round\ down)\ \text{or}$$

$$The\ position\ of\ Q_3 = \frac{3n}{4} = \frac{3\times 20}{4} = 15,\ n = 20\ (even)$$

$$The\ value\ of\ Q_3 = 51$$

Method 2 (percentile method)

Q_1: $\frac{25}{100} \times 20 = 5\ is\ an\ integer \longrightarrow \frac{5^{th}+6^{th}}{2}$ i.e. average of 5th and 6th

$$Q_1 = \frac{33+35}{2} = 34$$

Q_2: $\frac{50}{100} \times 20 = 10\ is\ an\ integer \longrightarrow \frac{10^{th}+11^{th}}{2}$ i.e. average of 10th and 11th observations

$$Q_2 = \frac{44+45}{2} = 44.5$$

Q_3: $\frac{75}{100} \times 20 = 15 \longrightarrow \frac{15^{th}+16^{th}}{2}$

$$Q_3 = \frac{51+53}{2} = 52$$

$$\begin{aligned} IQR &= Q_3 - Q_1 \\ &= 51 - 33 \\ &= 18 \end{aligned}$$

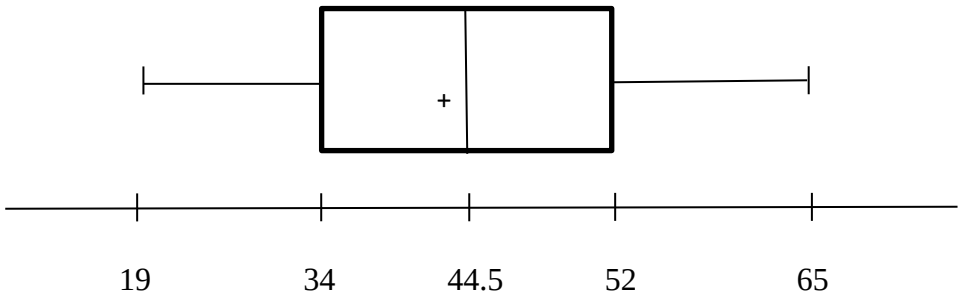
$$\begin{aligned} Q_1 - 1.5IQR &= 33 - 1.5(18) \\ &= 6 \end{aligned}$$

$$\begin{aligned} Q_3 + 1.5IQR &= 51 + 1.5(18) \\ &= 78 \end{aligned}$$

There is no observation below 18 and no observation above 78.

∴ there are no outliers

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i = \frac{1}{n}(x_1 + x_2 + \cdots + x_n)$$
$$= \frac{1}{20}(863) = 43.15$$



2. Construct a boxplot for the following data:

   3 7 2 14 4 29 3 9 1 20 10 7 2 42 3 5 12

   SOLUTION

   We can order the observations using a stem and leaf plot as follows:

0	3 7 2 4 3 9 1 7 2 3 5			0	1 2 2 3 3 3 4 5 7 7 9
1	4 0 2			1	0 2 4
2	9 0		→	2	0 9
3				3	
4	2			4	2

Method 1 (direct method)

1 2 2 3 (3) 3 4 5 (7) 7 9 10 (12) 14 20 29 42

Q_1 Q_2 Q_3

Method 2 (percentile method)

1 2 2 3 3 3 4 5 7 7 9 10 12 14 20 29 42

Q1: $\frac{25}{100} \times 17 = 4.25$ non-integer $\rightarrow 5^{th}$ i.e. 5th observation

$$Q_1 = 3$$

Q2: $\frac{50}{100} \times 17 = 8.5 \rightarrow 9^{th}$

$$Q_2 = 7$$

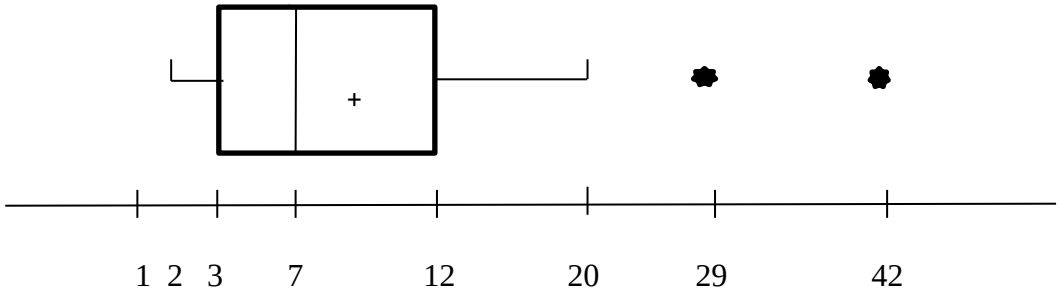
Q_3: $\frac{75}{100} \times 17 = 12.75 \rightarrow 13^{th}$

$$Q_3 = 12$$

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i = \frac{173}{17}$$
$$= 10.18$$
$$IQR = Q_3 - Q_1$$
$$= 12 - 3$$
$$= 9$$
$$Q_1 - 1.5IQR = 3 - 1.5(9)$$
$$= -10.5$$
$$Q_3 + 1.5IQR = 12 + 1.5(9)$$
$$= 25.5$$

There is no observation below -10.5

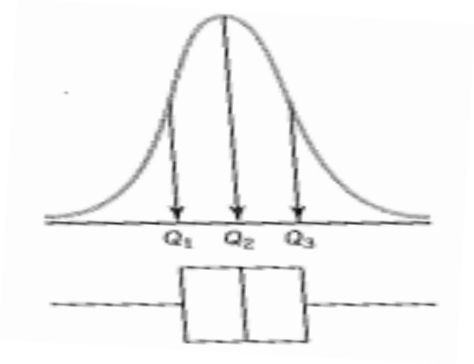
There are two observations above 25.5,(i.e., 29 and 42) these are outliers.



✓ Note

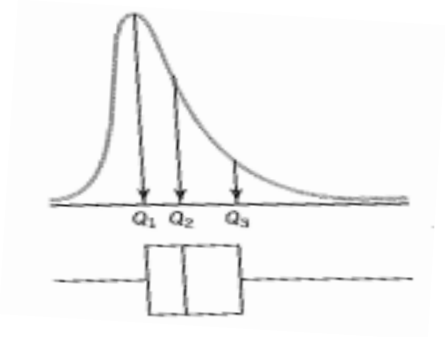
1. There are so many methods of finding the quartiles, however, each method would provide an approximate value for the quartiles. The simplest method is to find the median of the dataset, then the lower quartile becomes the median of the dataset which is less than or equal to the median and the upper quartile becomes the median of the data set which is greater or equal to the median.
2. To find the m[th] percentile, P_j, we first find its location in the ranked list of n data values. Evaluate $\frac{j}{n} \times n$ by considering the following:
   (a) if $\frac{j}{n} \times n$ is an integer, then the location of P_j is halfway between the data value in position $\frac{j}{n} \times n$ and the data value in the next position i.e. the average of these two values.
   (b) if $\frac{j}{n} \times n$ is not an integer, then the location of P_j is the next higher integer. P_j is the data value in this location.

3. A box plot for a symmetrical distribution would look like this:



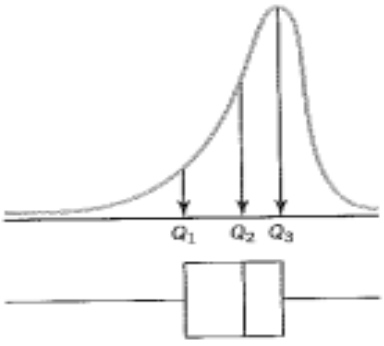
Here the whiskers are of equal length and the median is the middle of the box.

4. A box plot for a positively skewed or right skewed distribution would look like this:



Here the right-hand whisker is larger than the left-hand whisker and the median is nearer to the lower quartile.

5. A box plot for a negatively skewed or left skewed distribution would look like this:



Here the left hand whisker is larger than the right hand whisker and the median is nearer to the upper quartile.

4. MEASURES OF ASSOCIATION

- ✓ Often a manager or any decision maker is interested in the relationship between two variables.
- ✓ Thus, covariance and pearson product moment correlation coefficient are descriptive measures of the relationship between two variables.

4.1. COVARIANCE

- ✓ For a sample of size n with the observations (x_1, y_1) , (x_2, y_2), and so on, the sample covariance is defined as follows:

$$S_{xy} = \frac{\sum(x_i - \bar{x})\,(y_i - \bar{y})}{n-1} = \frac{\sum_{i=1}^{n} x_i y_i}{n} - \bar{x}\bar{y} = \sum_{i=1}^{n} x_i y_i - \frac{\sum x_i \sum y}{n}$$

- ✓ The Population covariance is given by

$$\sigma_{XY} = \frac{\sum(X_i - \mu_X)\,(Y_i - \mu_Y)}{N} = \frac{\sum_{i=1}^{N} X_i Y_i}{N} - \mu_X \mu_Y$$

where,

$S_{xy} =$ sample covariance,

$\bar{x} = sample\ mean\ of\ x$

$\bar{y} =$ sample mean of y

$\mu_X = population\ mean\ of\ X$

$\mu_Y = population\ mean\ of\ Y$

$\sigma_{XY} = population\ covariance.$

4.1.1. INTERPRETING COVARIANCE

(1) A positive value for S_{xy} indicates a positive linear association between *x* and *y*.

(2) A negative value for S_{xy} indicates a negative linear association between *x* and *y*.

(3) If the value of S_{xy} is close to zero, indicating no linear association between *x* and *y*.

4.2. PEARSON PRODUCT MOMENT CORRELATION COEFFICIENT

- ✓ The product moment correlation coefficient, r, is the numerical value between -1 and 1 inclusive which indicates the linear degree of scatter.
- ✓ For sample data, the Pearson product moment correlation coefficient is defined as follows:

$$r_{xy} = \frac{S_{xy}}{S_x S_y}$$

where

$$S_x = \sqrt{S_{xx}} = \sqrt{\frac{\sum(x_i - \bar{x})^2}{n-1}} = \sqrt{\frac{\sum x^2}{n} - \bar{x}^2} = \sqrt{\sum x^2 - \frac{(\sum x)^2}{n}}$$

$$S_y = \sqrt{S_{yy}} = \sqrt{\frac{\sum(y_i - \bar{y})^2}{n-1}} = \sqrt{\frac{\sum y^2}{n} - \bar{y}^2} = \sqrt{\sum y^2 - \frac{(\sum y)^2}{n}}$$

- ✓ For population data, Pearson Product Moment Correlation Coefficient is defined as follows:

$$\rho_{XY} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y}$$

where,

$r_{XY} = sample\ correlation\ coefficient$

$\rho_{XY} = population\ correlation\ coefficient$

EXAMPLE

The following table shows the marks of ten candidates in ISB 140 and DS 110 for the year 2019. Find the covariance and the pearson product moment correlation coefficient and comment on the value.

Mark in ISB 140 (x)	18	20	30	40	46	54	60	80	88	92
Mark in DS 110 (y)	42	54	60	54	62	66	80	66	80	100

SOLUTION

There are ten pairs of data, $n = 10$

x	y	x^2	y^2	xy
18	42	324	1764	756
20	54	400	2916	1080
30	60	900	3600	1800
40	54	1600	2916	2160
46	62	2116	3844	2852
54	68	2916	4624	3672
60	80	3600	6400	4800
80	66	6400	4356	5280
88	80	7744	6400	7040
92	100	8464	10 000	9200
$\sum x = 528$	$\sum y = 666$	$\sum x^2 = 34464$	$\sum y^2 = 46820$	$\sum xy = 38640$

$$\bar{x} = \frac{\sum x}{n} = \frac{528}{10} = 52.8 \text{ and } \bar{y} = \frac{\sum y}{n} = \frac{666}{10} = 66.6$$

$$Covariance\text{: } S_{xy} = \frac{\sum_{i=1}^{n} x_i y_i}{n} - \bar{x}\bar{y} = \frac{38640}{10} - 52.8 \times 66.6 = 347.52$$

$$\Leftrightarrow S_{xy} = \sum_{i=1}^{n} x_i y_i - \frac{\sum x_i \sum y}{n} = 38640 - \frac{52.8 \times 66.6}{10} = 3475.2$$

$$S_{xx} = \frac{\sum x^2}{n} - \bar{x}^2 = \frac{34464}{10} - 52.8^2 = 658.56$$

$$\Leftrightarrow S_{xx} = \sum x^2 - \frac{(\sum x)^2}{n} = 34464 - \frac{528^2}{10} = 6585.6$$

$$S_{yy} = \frac{\sum y^2}{n} - \bar{y}^2 = \frac{46820}{10} - 66.6^2 = 246.44$$

$$\Leftrightarrow S_{yy} = \sum y^2 - \frac{(\sum y)^2}{n} = 34464 - \frac{528^2}{10} = 2464.4$$

Thus, the pearson product moment correlation coefficient is

$$r_{xy} = \frac{S_{xy}}{S_x S_y} = \frac{347.52}{\sqrt{658.56} \times \sqrt{246.44}} = 0.8626$$

$$\Leftrightarrow r_{xy} = \frac{S_{xy}}{S_x S_y} = \frac{3475.2}{\sqrt{6585.6} \times \sqrt{2464.4}} = 0.8626 \ldots$$

The product moment correlation coefficient is 0.86 (2 s.f.), indicating good positive correlation.